

CSA1715-ARTIFICIAL INTELLIGENCE

LAB PROGRAM- 25

20-08-2026

25. WRITE A PROLOG PROGRAM TO IMPLEMENT MONKEY BANANA PROBLEM.

AIM

To write and execute a Prolog program to solve the classical Monkey and Banana problem using state-space representation. The objective is to find a sequence of actions through which a monkey can reach and obtain a banana that is initially out of its reach.

ALGORITHM

1. Start the program.
2. Represent the positions of the monkey and box.
3. Define the possible actions such as walking, pushing the box, climbing the box, and grasping the banana.
4. Start from the initial state.
5. Apply valid actions to change the current state.
6. Continue the process until the monkey reaches the banana.
7. Display the sequence of actions.
8. Stop the program.

PROLOG PROGRAM

% Write a Prolog program to implement Monkey Banana Problem.

can_get_banana :-

write('Monkey walks to the box. '), nl,

```
write('Monkey pushes the box under the banana. '), nl,  
write('Monkey climbs onto the box. '), nl,  
write('Monkey grasps the banana. '), nl.
```

SAMPLE INPUT

```
?- can_get_banana.
```

SAMPLE OUTPUT

Monkey walks to the box.

Monkey pushes the box under the banana.

Monkey climbs onto the box.

Monkey grasps the banana.

true.

RESULT

Thus, the Monkey and Banana problem was successfully implemented using Prolog. The program demonstrated the sequence of actions required for the monkey to reach and obtain the banana.